

Ayush Sharma

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EDUCATION

New York University

Manhattan, NY

Masters of Science in Computer Science | **CGPA: 3.9/4.0**

Aug 2025 – May 2027

Coursework: Big Data, Computer Networks, Operating Systems, Algorithms, Machine Learning, Databases

Vellore Institute of Technology

Vellore, TN

Bachelor of Technology in Computer Science | **CGPA: 3.9/4.0**

Jul 2019 – Jun 2023

Coursework: Data Structures & Algorithms, Artificial Intelligence, Computer Vision, Cybersecurity

TECHNICAL SKILLS, AWARDS, CERTIFICATIONS, OPEN-SOURCE

Languages: Java, Python, C/C++, Golang, SQL/NoSQL, MATLAB

Frameworks & Tools: Docker, Tensorflow, REST APIs, Git, Terraform, Linux, Agile, Hadoop, Spark, GCP, Azure

Awards: Winner at HackNYU, Startup Week Buildathon, Daytona HackSprint, Elevenlabs Worldwide Hack, Pulse NYC

Certifications: Certified Kubernetes Application Developer (CKAD), AWS Certified Cloud Practitioner

Open-Source: Google Summer of Code (GSoC) 2026 – Ceph, a distributed object, block, and file storage platform

Paper: Sharma, A. (2020). A Comparative Study on Efficiencies of Variants of CNNs on Image Classification. [\[Paper\]](#)

EXPERIENCE

Software Engineer 1

Jan 2023 – Jul 2025

Hewlett Packard Enterprise

Bengaluru, KA

- Maintained carrier-grade media server serving **100M+ users with 99.999% uptime**, handling SIP signaling, RTP media, and real-time speech recognition/synthesis
- Architected & **Patented** on-device LLM inference pipeline on Gemini Nano with context-aware NPU gating for real-time scam detection across sentences. **Presented at Mobile World Congress 2024**
- Migrated 3 VM-based services to Kubernetes, **improving deployment speed and horizontal scalability**.
- Built distroless images eliminating shells, package managers, and unused OS utilities, **cutting storage by 75%**.
- Built a dependency tracker adopted by dev, QA, release teams, closing audit gaps and **cutting CVEs by 90%**.
- Orchestrated data aggregation with **monitoring and alerting of 100+ metrics** using Prometheus and Grafana
- Built license generation tool **adopted by 8 teams**. Supports tamper-proof verification, Fluentd event monitoring
- Reduced QA regression cycle from **5 days to 1 hour** by integrating Selenium for GUI feature testing into Jenkins CI/CD pipelines, and hardened application security by embedding penetration tests.

Machine Learning Research Intern

Feb 2026 – April 2026

Courant Institute of Mathematical Sciences | CILVR Lab | Advisor: Prof. Pavel Izmailov

New York, NY

- Built SVG generation model on 89K files achieving **85% XML validity, 78% render rate, 200ms p95 latency**
- Trained Transformers from 1M to 88M params on 100M+ tokens to study scaling laws. Implemented zero-shot hyperparameter transfer, **saving 4-5 tuning runs and 10+ GPU hours** per scale

Software Engineer Intern

Oct 2021 – April 2022

Samsung R&D Institute

Bengaluru, KA

- Developed a C++ library abstracting 6 protocols (TCP,UDP,MQTT,gRPC,HTTP,FTP) for mesh communication.
- Achieved **10% lower latency and 15% higher throughput** over TCP baseline through protocol-specific optimizations like multi-threading, async I/O, connection pooling, ProtoBuf serialization, and zero-copy.

PROJECTS

Social Recommendation Engine | *FastAPI, NetworkX, Stripe, GPT-5, LangChain*

[\[Project\]](#)

- Built recommendation engine using collaborative filtering, temporal, spatial & social graph analysis for venue/companion matching across 100k+ check-ins, with multi AI agents automating reservations & payments.

LeetCourt | *React, ElevenLabs, OpenRouter (GPT-5), Daytona sandbox, MongoDB*

[\[Project\]](#)

- Led a team & developed voice-interactive courtroom simulator with multi AI agents - judge & counsel, with reactive scoring engine that gamifies legal training by transforming PDF case files into real-time adversarial debates.

False Markets | *FastAPI, TypeScript, Next.js, Redis, PostgreSQL, CoinGecko, yfinance, Alpaca*

[\[Project\]](#)

- Built evolutionary trading system on 1-min tick data across 25+ crypto assets - agents carrying encoded genomes (RSI/MACD weights, Kelly criterion, trailing stops), evolving via fitness scores (Sharpe ratio, drawdown, win rate)